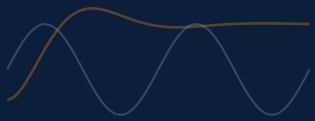
WEEK 2 · DAY 4 / 5

Mechatronics and Robotics Engineering (50 h)



Flexible Automation, Safety and Robot Sensors

5.6 Material handling & HRC ▪ 5.7 Safety & standards ▪ 6.1 Measurement principles ▪ 6.2 Position & motion sensors ▪ 6.3 Force, proximity & range sensors

Mechatronics and Robotics Engineering Programme

Week 2 · Day 4 ▪ 2 weeks · 10 days · 50 hours

Agenda and learning outcomes

Code	Session	Time
5.6	Material handling and flexible systems	60 min
5.7	Safety and standards	60 min
6.1	Sensors and measurement principles	60 min
6.2	Position and motion sensors	60 min
6.3	Force/torque, tactile, proximity and range sensors	60 min

5 h net teaching time; 10-min breaks roughly every hour are extra.

By the end of today you can . . .

- size a conveyor and a small AMR fleet
- name the four collaborative modes
- walk through an ISO 12100 risk assessment
- calculate a light-curtain safety distance
- tell accuracy, precision and resolution apart
- explain how encoders and IMUs measure motion
- choose a sensor for force, presence or distance

What we learned so far: the robot work-cell (W2D3)

5.2 Applications: welding, painting, palletising, tending

5.3 Work-cell design: layout, reach, cycle time, EOAT

5.4 PLCs: scan cycle, I/O, ladder logic, sequences

5.5 Robot programming: frames, teach pendant, OLP

Today Moving parts between cells, keeping people safe, and the sensors that make it all work

Three ideas we reuse today

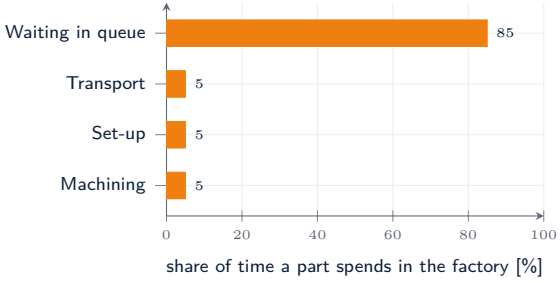
- **Cycle time**: time to finish one part; the slowest station sets the pace.
- **PLC I/O**: sensors are the PLC's inputs, actuators its outputs.
- **Frames**: base, tool and user frames describe *where* things are.

Yesterday we built one cell. Today we connect cells, add people, and ask: how does the robot *know* where things are?

Session 1

5.6 Material handling and flexible systems

Why material flow matters



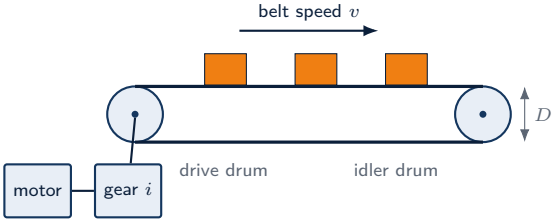
Illustrative split for a traditional job shop.

Analogy: airport baggage

Your suitcase spends minutes in the plane's hold and hours waiting on belts and trolleys. Factories are the same: parts mostly *wait*.

- **Material handling**: moving, storing and buffering parts between operations.
- A fast robot is wasted if parts arrive late.
- Flexible handling lets one line make many product variants.

Conveyors: the factory's moving pavement



- **Belt**, **roller** and **accumulating** conveyors; indexing tables.
- **Accumulation** lets parts queue without pushing each other.

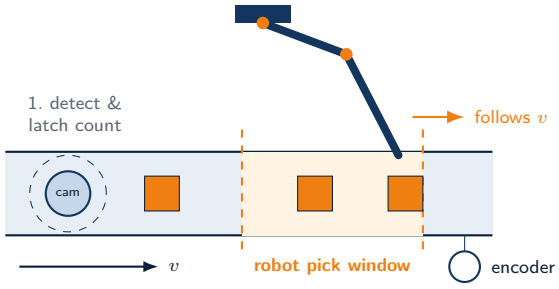
$$v = \pi \, D \, \frac{n_{\text{motor}}}{i}$$

v belt speed [m/s], D drum diameter [m], n speed [rev/s], i gear ratio.

Worked example: belt speed

1. Motor 1500 rpm $= 25$ rev/s, gear $i = 25$
2. Drum: $25/25 = 1$ rev/s
3. $D = 0.1$ m: $v = \pi \cdot 0.1 \cdot 1 \approx 0.31$ m/s
4. Parts every 0.5 m $\Rightarrow 0.31/0.5 \approx 0.63$ parts/s ≈ 37 parts/min

Conveyor tracking: picking from a moving belt



1. A sensor or camera sees the part and **latches** the conveyor encoder count.
2. Robot computes where the part is now: $x = x_0 + \Delta(\text{counts}) \cdot s$.
3. Robot moves *with* the belt inside the pick window.

Worked example: counts to mm

1. Encoder: 4000 counts per drum rev
2. Drum circumference $\pi \cdot 100\,\text{mm} \approx 314\,\text{mm}$
3. $s = 314/4000 \approx 0.079\,\text{mm}$ per count

AGV and AMR: vehicles that move material



Industrial robot and AGV, Marta Veverica, CC BY-SA 4.0, via Wikimedia Commons

	AGV	AMR
Guidance	wire, magnetic tape, QR floor codes	natural features (LiDAR map)
Obstacle	stops and waits	plans a path around it
Layout change	re-lay the tape	edit the map
Typical use	fixed, heavy, repetitive routes	changing routes, shared spaces

- **AGV**: automated guided vehicle (a train on invisible rails).
- **AMR**: autonomous mobile robot (a taxi that finds its own way).

Going further: AMRs localise with SLAM (simultaneous localisation and mapping) and drive with differential-drive kinematics (4.7).

Worked example: how many AMRs do we need?



speed 1 m/s, 60 s to load or unload

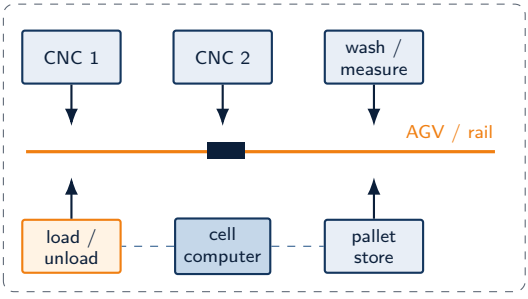
Rule of thumb: plan 15–25 % spare capacity for charging, traffic and peaks.

Step by step

1. Demand: **60 transports per hour**
2. Travel: $2 \cdot 200/1 = 400$ s; handling $2 \cdot 60 = 120$ s
3. Round trip $= 520$ s ≈ 8.7 min $\Rightarrow 60/8.7 \approx 6.9$ trips/h per AMR
4. $60/6.9 \approx 8.7 \Rightarrow$ 9 AMRs; with 80 % availability $9/0.8 \approx 11.3 \Rightarrow$ **12 AMRs**

$$N_{\text{AMR}} = \frac{\text{demand [trips/h]} \cdot t_{\text{round trip}}\,[\text{h}]}{\text{availability}}$$

Flexible manufacturing system (FMS) cell

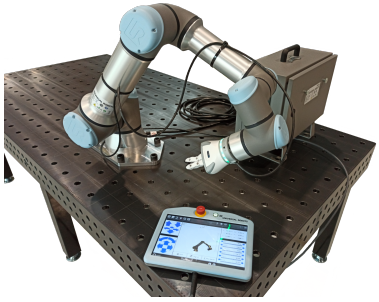


- **FMS**: several CNC machines + automated transport + a computer that schedules jobs.
- Parts ride on **pallets** with fixtures; any machine can take any pallet.
- Robots load machines; AGVs or rail carts move pallets.

Bottleneck example

Stations take 40, 60 and 45 s per part. The **slowest** sets the pace: one part per 60 s = 60 parts/h.

Collaborative robots (cobots)



UR16e robot arm, Auledas, CC BY-SA 4.0, via Wikimedia Commons

Definition

A **cobot** is a robot designed to work *in the same space* as people, without a fence, when the risk assessment allows it.

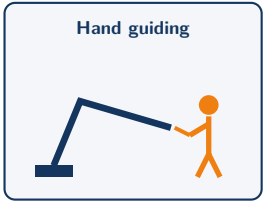
- Light arms, rounded shapes, no pinch points.
- Joint torque or current sensing detects a collision and stops.
- Speed and force are limited by safety settings.
- Easy to teach by hand (hand guiding).

A cobot with a knife in its gripper is **not** safe. The *application* is collaborative, not the robot alone.

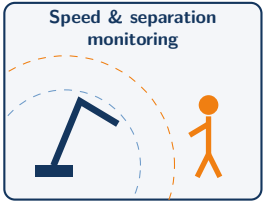
Human–robot collaboration: four modes



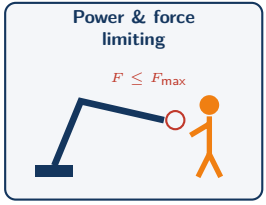
robot stops while the person is in the shared space



person moves the robot with an enabling handle



robot slows or stops as the person comes closer



contact allowed, but forces stay below body limits

- Defined in **ISO 10218** (robots & robot systems); **ISO/TS 15066** gives body-region force and pressure limits for contact.
- Most real cells combine modes, e.g. speed & separation plus a monitored stop.

Going further: in power & force limiting, the energy of a contact $\frac{1}{2}\mu v^2$ must stay below what the body region can absorb, $F_{\max}^2/(2k)$, which caps the relative speed: $v_{\max} = F_{\max}/\sqrt{\mu k}$.

Check your understanding: material handling

Check your understanding

1. A conveyor drum of 0.2 m diameter turns at 0.5 rev/s. What is the belt speed?
2. An AGV meets a box left in the aisle. What does it do, and what would an AMR do?
3. A cobot carries a sharp deburring tool. Can the cell run without any safeguards? Why?

Summary and key terms: material handling

- Parts spend most time waiting: flow matters as much as speed.
- Conveyor speed $v = \pi D n / i$; tracking uses a belt encoder.
- AGVs follow fixed guides; AMRs navigate with a map.
- FMS: machines + transport + scheduling computer.
- Cobots: four collaborative modes, judged per application.

Key terms

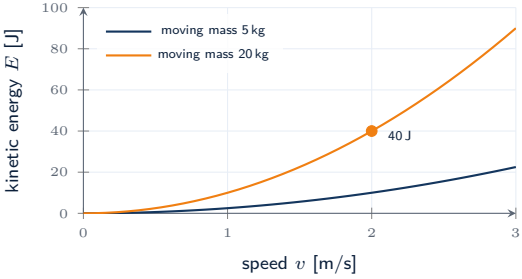
accumulation parts queue on the belt
conveyor tracking robot follows the belt
AGV / AMR guided / autonomous vehicle
FMS flexible manufacturing system
bottleneck slowest station
HRC human–robot collaboration

[Groover, 2019]; ISO 10218-2; ISO/TS 15066.

Session 2

5.7 Safety and standards

Why robots are dangerous



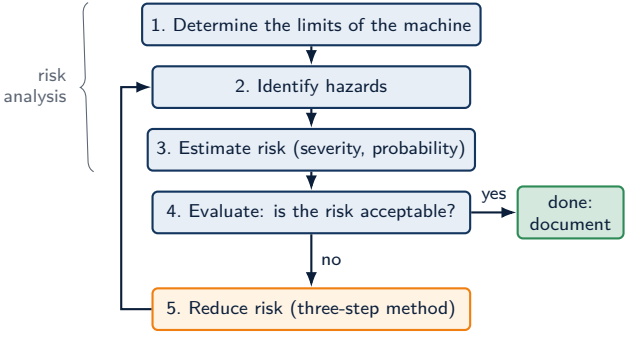
- Hazards: **impact**, **crushing** and **trapping**, flying parts, the tool itself (laser, weld arc, blade).
- Robots move *fast*, *without warning* and may start unexpectedly.

$$E = \tfrac{1}{2}\, m\, v^2$$

Worked example

1. Effective moving mass 20 kg at 2 m/s
2. $E = 0.5 \cdot 20 \cdot 2^2 = 40$ J
3. Like a 4 kg brick dropped from 1 m: enough to break bones.

ISO 12100: the risk assessment loop

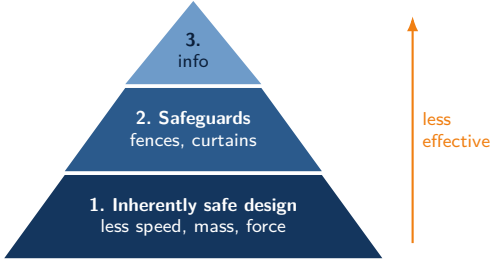


Definition

Risk = combination of the **severity** of a possible harm and the **probability** that it occurs.

- ISO 12100 is the basic (type A) standard for *all* machinery.
- It is a loop: after each safeguard, check again for new hazards.
- The result is written down: the risk assessment file.

Three-step risk reduction and the standards family



Step 3 = warnings, signs, training, PPE.

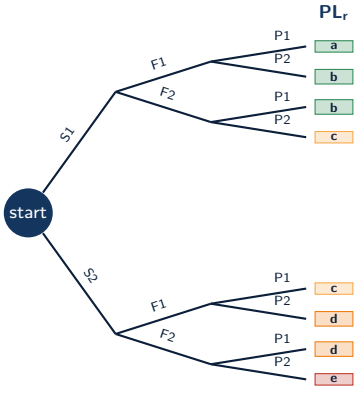
Type A ISO 12100: risk assessment

Type B ISO 13849, IEC 62061 (safety controls); ISO 13855 (distances)

Type C ISO 10218-1/-2 (industrial robots); ISO/TS 15066 (collaboration)

- A: general, B: generic safety aspects, C: one machine type.
- A type C standard takes priority for its machine.

How strong must a safety function be? The risk graph



S	severity: S1 slight, S2 serious
F	exposure: F1 rare, F2 frequent
P	avoidance: P1 possible, P2 hardly

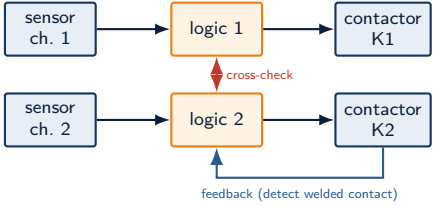
- Result: the **required performance level** PL_r, from a (weak) to e (strong).
- Robot safety functions are typically required at **PL d**.

Example

Operator reaches into a robot cell several times per hour (F2), injury could be serious (S2), robot is fast (P2) ⇒ **PL e**.

Performance level and redundant (two-channel) design

Category 3: one fault does not lose the safety function

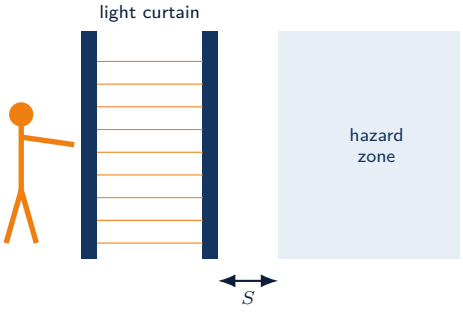


PL	dangerous failures per hour
a	10^{-5} to 10^{-4}
b	$3 \cdot 10^{-6}$ to 10^{-5}
c	10^{-6} to $3 \cdot 10^{-6}$
d	10^{-7} to 10^{-6}
e	10^{-8} to 10^{-7}

- PL d: at most one dangerous failure in about **100 years** of operation (10^{-6}/h $\approx$ 1 per 114 years).
- Reached by **redundancy** + **diagnostics** + reliable parts.

Going further: ISO 13849-1 combines category, $MTTF_D$, diagnostic coverage and common-cause failure.

Light curtains and the safety distance



- Beams of infrared light; any blocked beam $\Rightarrow$ stop signal.
- **Detection capability** d: 14 mm finger, 30 mm hand.

$$S = K \cdot T + C \qquad \text{(ISO 13855)}$$

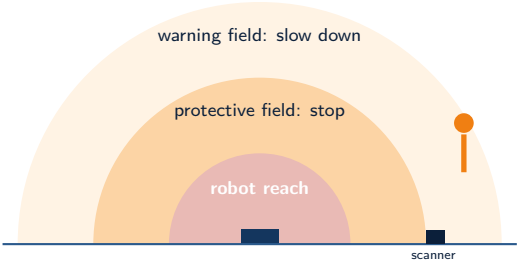
S min. distance [mm], K approach speed (2000 mm/s for a hand), T total stopping time [s], C intrusion allowance $= 8(d - 14)$ [mm].

Worked example

1. Curtain 20 ms + robot stop 180 ms $\Rightarrow T = 0.2$ s
2. $d = 14$ mm $\Rightarrow C = 8 \cdot (14 - 14) = 0$
3. $S = 2000 \cdot 0.2 + 0 = 400$ mm

A slower-stopping robot needs the curtain **further away**: measure the real stopping time!

Laser scanners and speed & separation monitoring

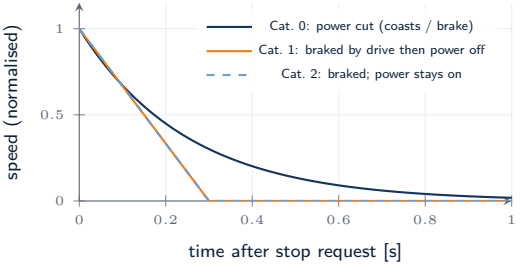


- A **safety laser scanner** measures distance by light time-of-flight in a fan (Session 6.3).
- Configurable **fields**: warning ⇒ reduced speed; protective ⇒ stop.
- **Speed & separation monitoring**: the allowed robot speed depends on how far away the person is.

Intuition

Like a car's adaptive cruise control: the closer the car ahead, the slower you go, and you must always be able to stop in time.

Emergency stop, stop categories and the safety PLC



Stop categories of IEC 60204-1 (illustration).

- **E-stop**: red mushroom button on yellow; latching, two channels; usually Cat. 0 or 1.
- **Safety PLC**: redundant processors, self-tests, certified for PL e; reads curtains, door switches, scanners.
- **Safe torque off (STO)**: the drive removes torque without opening the motor cable.

> Standard PLC: run the process. Safety PLC: decide whether the process may run at all.

Check your understanding: safety

Check your understanding

1. List the three steps of ISO 12100 risk reduction in order of preference.
2. A light curtain sees 14 mm objects; total stopping time is 0.15 s. What is the minimum safety distance?
3. Why does a safety function use two channels instead of one very good channel?

Summary and key terms: safety and standards

- ISO 12100: limits → hazards → estimate → evaluate → reduce, repeat.
- Prefer safe design, then safeguards, then information.
- Risk graph gives PL_r; redundancy + diagnostics reach it.
- Safety distance $S = KT + C$; stopping time matters.
- Robot standards: ISO 10218, ISO/TS 15066.

Key terms

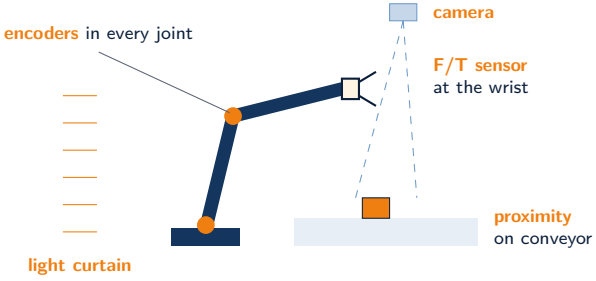
risk severity × probability of harm
PL performance level a–e
category structure (e.g. two channels)
safeguard guard or protective device
STO safe torque off
protective field zone that triggers a stop

ISO 12100; ISO 13849-1; ISO 13855; IEC 60204-1.

Session 3

6.1 Sensors and measurement principles

The senses of a robot cell

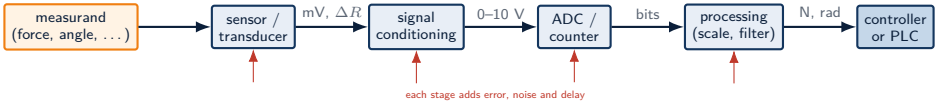


- **Proprioceptive**: the robot senses *itself* (joint angles, motor current).
- **Exteroceptive**: senses the *world* (parts, people, forces).
- A typical cell has dozens of sensors; each one is an input to a controller or a PLC.

Analogy

Close your eyes and touch your nose: that is proprioception. Seeing the cup you reach for is exteroception.

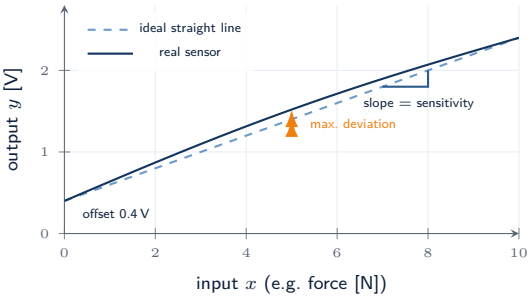
The measurement chain



- **Transducer**: converts one physical quantity into another (strain → resistance).
- **Signal conditioning**: amplify, filter, isolate, linearise.

> The chain is only as good as its weakest link: a perfect sensor with a noisy cable gives a noisy measurement. Sampling and filtering come on W2D5 (6.5, 6.6).

Static characteristic: sensitivity, offset, range, linearity



$$y = y_0 + S\,x$$

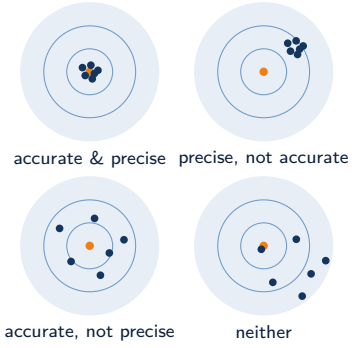
S **sensitivity** [V/N], y_0 **offset** [V].

- **Range**: smallest to largest input, e.g. 0–10 N; **full scale** (FS) = 10 N.
- **Linearity error**: largest gap to the straight line, in % of FS.

Example

Max. gap 0.12 V on an output span of 2.0 V $\Rightarrow$ linearity error $0.12/2.0 = 6\,\%$ FS.

Accuracy, precision and resolution



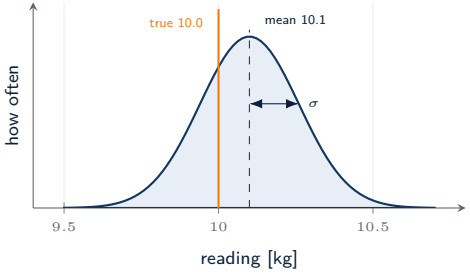
Refresher: the dartboard picture

Accuracy: how close the darts land to the bull's-eye (true value).
Precision (repeatability): how tightly they cluster.
Resolution: the smallest change the sensor can show.

Robots too

A typical industrial robot repeats a taught point to about $\pm 0.05\,\text{mm}$ (precision) but reaches a *computed* point only to a few tenths of a mm (accuracy) [ISO 9283].

Random and systematic errors



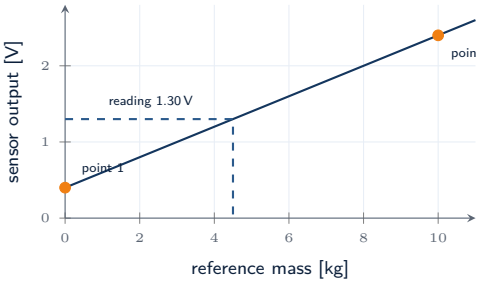
- **Systematic error** (bias): mean − true; fixed by **calibration**.
- **Random error**: spread σ; reduced by **averaging**.

$$\bar{x} = \frac{1}{n}\sum x_i, \quad \sigma = \sqrt{\frac{1}{n-1}\sum (x_i - \bar{x})^2}$$

Worked example: five readings

1. Readings: 10.2, 10.0, 10.1, 9.9, 10.3 kg
2. Mean $\bar{x} = 50.5/5 = 10.1$ kg $\Rightarrow$ bias $+0.1$ kg
3. Deviations: $0.1, -0.1, 0, -0.2, 0.2$; squares sum 0.10
4. $\sigma = \sqrt{0.10/4} \approx 0.16$ kg

Calibration: making the numbers right



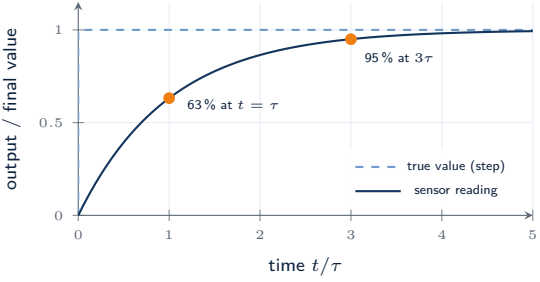
- Apply known inputs (**reference standards**), record outputs, fit a line.
- Then invert the line to turn volts into kilograms.

Worked example: two-point calibration

1. 0 kg $\rightarrow$ 0.40 V; 10 kg $\rightarrow$ 2.40 V
2. Sensitivity $S = (2.40 - 0.40)/10 = 0.20\,\text{V/kg}$
3. Offset $y_0 = 0.40\,\text{V}$
4. Reading 1.30 V: $x = (1.30 - 0.40)/0.20 = 4.5\,\text{kg}$

Going further: in robotics, "calibration" also means identifying the robot's true link lengths, camera parameters or tool frame (W2D5).

Dynamic behaviour: sensors are not instant



Analogy: a kitchen thermometer

Put it in hot soup: the reading climbs for a while before it settles. The sensor needs time to "catch up".

$$y(t) = y_\infty \left(1 - e^{-t/\tau}\right)$$

τ **time constant** [s]. Same first-order response as W1D2 (2.2).

Example

$\tau = 2\,\text{ms} \Rightarrow 95\,\%$ after $3\tau = 6\,\text{ms}$; a 1 kHz control loop sees the change only after six cycles.

Check your understanding: measurement principles

Check your understanding

1. A scale shows 10.2, 10.2, 10.3 kg for a true 10.0 kg mass. Is it accurate, precise, both or neither?
2. A sensor reads 1.0 V at 0 N and 3.0 V at 100 N. What force gives 2.2 V?
3. A temperature probe has $\tau = 4$ s. How long until it shows 95 % of a sudden change?

Summary and key terms: measurement principles

- Measurement chain: sensor $\rightarrow$ conditioning $\rightarrow$ ADC $\rightarrow$ processing.
- Static: sensitivity, offset, range, linearity.
- Accuracy $\neq$ precision $\neq$ resolution.
- Bias $\rightarrow$ calibrate; noise $\rightarrow$ average.
- Dynamic: time constant τ, 95 % after 3τ.

Key terms

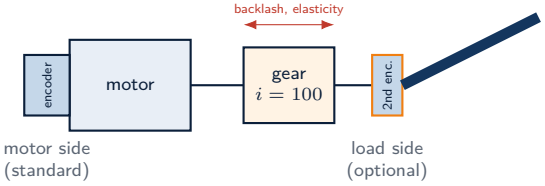
transducer converts one quantity to another
sensitivity output change per input change
full scale largest rated input
bias systematic error
repeatability precision of repeated readings
time constant 63 % response time

[Fraden, 2016]; [Bentley, 2005]; ISO 9283.

Session 4

6.2 Position and motion sensors

Where position sensors sit in a robot joint



- Every robot joint has a sensor on the **motor shaft**; the gear multiplies its resolution.
- But the gear's backlash and twist are *not* seen by a motor-side sensor.
- High-accuracy robots add a **secondary (load-side) encoder**.

Worked example: gear helps resolution

1. Motor encoder: $2^{17} = 131\,072$ counts/rev
2. Gear $i = 100$: 13.1 million counts per joint rev
3. $360^\circ / 13.1 \cdot 10^6 \approx 0.00003^\circ$ per count

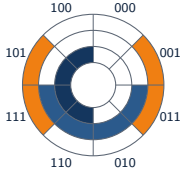
Today: encoders, resolvers and inertial sensors (IMU). Cameras: W2D5 (6.4).

Refresher: binary numbers and Gray code

Refresher: binary

Each bit is a power of 2: $101_2 = 1{\cdot}4 + 0{\cdot}2 + 1{\cdot}1 = 5$.
N bits give 2^N values: 3 bits $\rightarrow$ 8, 12 bits $\rightarrow$ 4096, 17 bits $\rightarrow$ 131 072.

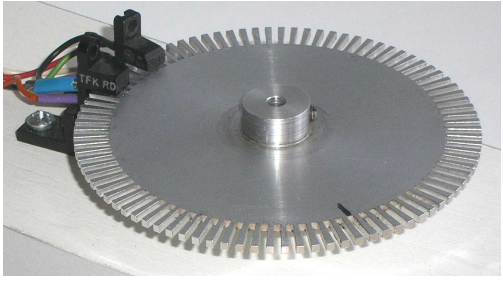
Value	Binary	Gray
2	010	011
3	**011**	010
4	**100**	110
5	101	111



3-bit Gray-code disk

- Binary $3 \rightarrow 4$: **all 3 bits** flip; if they flip at slightly different moments the reader may see a wrong value.
- **Gray code**: only **one bit** changes per step $\Rightarrow$ no false readings.

Optical incremental encoders



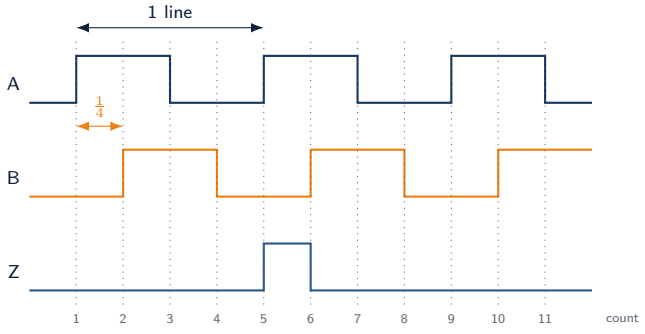
Encoder model, Rrudzik, CC BY-SA 4.0, via Wikimedia Commons

- An LED shines through a disk with N slits (lines) onto photodiodes.
- Outputs: square waves **A** and **B** (shifted by a quarter period) and an index pulse **Z** once per revolution.
- The drive counts pulses: position *relative* to switch-on $\Rightarrow$ **homing** needed.

Example: pulse frequency

1. $N = 2500$ lines, $n = 6000\,\text{rpm} = 100\,\text{rev/s}$
2. $f_A = N \cdot n = 2500 \cdot 100 = 250\,\text{kHz}$
3. The counter input must handle at least 250 kHz.

Quadrature signals and $\times 4$ counting

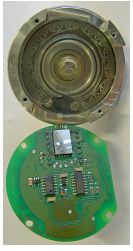


- Count **every edge** of A and B: 4 counts per line $\Rightarrow 4N$ counts/rev.
- **Direction**: if A changes before B we turn forward, otherwise backward.
- Both changing at once = a lost count $\Rightarrow$ error flag.

Example

$N = 2500$: $4N = 10\,000$ counts/rev $\Rightarrow 360^\circ/10\,000 = 0.036^\circ$ per count.

Absolute encoders: no homing after power-up



13-track Gray-code encoder, Mike1024, public domain, via Wikimedia Commons

- **Absolute**: every angle has its own code (many tracks, Gray-coded) $\Rightarrow$ position known at switch-on, **no homing**.
- **Single-turn**: e.g. 17–23 bit per rev. **Multi-turn**: also counts whole revolutions (gears or a battery-backed counter).
- Robots use multi-turn absolute encoders: after a power cut the arm knows its pose at once.

Protocol	Features
SSI	simple, one direction
BiSS-C, EnDat 2.2	two-way, CRC check, diagnostics

Resolver: a rotating transformer

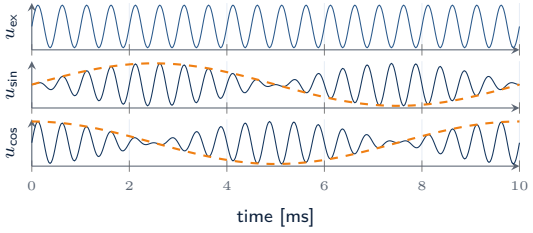
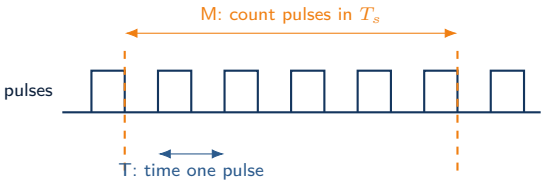


Illustration: carrier 2 kHz, rotor turning slowly.

- Rotor coil fed with a sine **carrier** u_{ex}.
- Two stator coils at 90° pick up $\sin\theta$ and $\cos\theta$ times the carrier.
- The angle is in the **envelopes** (dashed): $\theta = \mathrm{atan2}(\sin, \cos)$.
- No electronics in the motor: robust to heat, shock, vibration.

Going further: a **resolver-to-digital converter** (RDC) demodulates the signals and uses a small tracking loop to output angle and speed.

Speed from an encoder and the low-speed problem



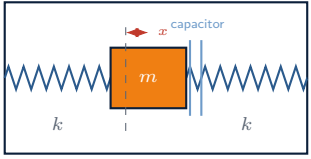
- **M-method**: count pulses in a fixed time T_s: good at high speed.
- **T-method**: time between two pulses: good at low speed.
- Modern drives combine both.

Worked example: M-method

1. $4N = 10\,000$ counts/rev, $T_s = 125\,\mu s$
2. One count per sample $= \dfrac{1}{10\,000 \cdot 125 \cdot 10^{-6}} = 0.8$ rev/s
3. $= 48$ rpm: the smallest speed step
4. At 60 rpm the reading jumps between 48 and 96 rpm!

Remedies: more counts/rev, longer windows, filtering (W2D5, 6.6).

Accelerometers: a mass on a spring





6-DoF IMU breakout (LSM6DS3), SparkFun, CC BY 2.0, via Wikimedia Commons

$$x = \frac{m}{k}\, a$$

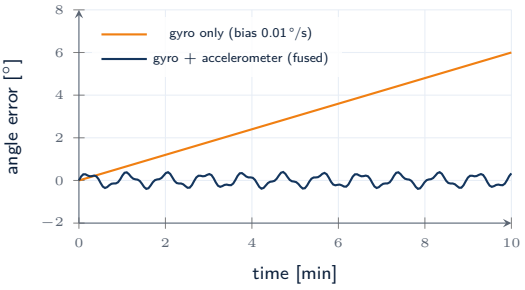
x deflection [m], m proof mass [kg], k spring stiffness [N/m], a acceleration [m/s^2].

- Silicon (**MEMS**) chip: a tiny mass moves, a capacitor measures x.
- At rest it measures **gravity**: $a_x = g \sin\theta$ gives tilt.

Worked example: tilt

1. At rest, $a_x = 4.9\,\text{m/s}^2$, $g = 9.81\,\text{m/s}^2$
2. $\sin\theta = 4.9/9.81 = 0.5 \Rightarrow \theta = 30^\circ$

Gyroscopes and the IMU: drift and fusion



- A **gyroscope** measures **angular rate** ω [°/s] (MEMS: vibrating mass + Coriolis force).
- Angle = integral of rate: a small **bias** grows into a large angle error (**drift**).
- **IMU** = 3-axis accelerometer + 3-axis gyro (sometimes magnetometer).

Worked example: drift

Bias $0.01\,°/s \times 600\,s$ (10 min) $= 6°$ error.

Going further: a complementary or Kalman filter trusts the gyro for fast changes and the accelerometer for the long-term tilt (W2D5, 6.6).

Check your understanding: position and motion sensors

Check your understanding

1. A 1000-line incremental encoder is decoded $\times 4$. How many counts per revolution, and how many degrees per count?
2. After a power failure, which encoder type needs a homing run: incremental or absolute? Why?
3. An accelerometer at rest reads $a_x = 9.81\,\mathrm{m/s^2}$. How is it tilted?

Summary and key terms: position and motion sensors

- Motor-side encoder in every joint; gear multiplies resolution.
- Incremental A/B/Z: $4N$ counts/rev, homing needed.
- Absolute (Gray code, multi-turn): pose known at power-up.
- Resolver: robust analogue sensor; angle from atan2.
- IMU: accelerometer (tilt) + gyro (rate, drifts).

Key terms

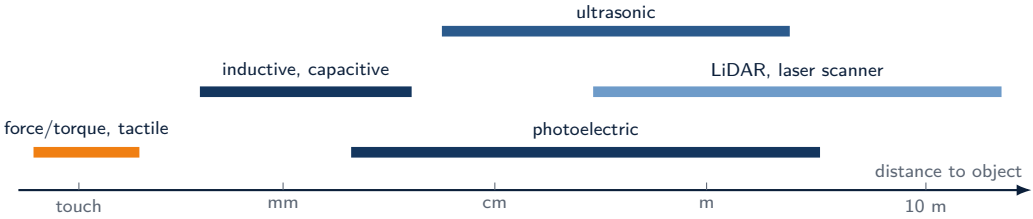
quadrature two signals 90° apart
homing finding a known zero position
Gray code one bit changes per step
resolver rotary transformer sensor
MEMS micro-machined silicon sensor
drift slowly growing error from bias

[Siciliano & Khatib, 2016]; [Fraden, 2016].

Session 5

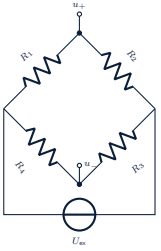
6.3 Force/torque, tactile, proximity and range sensors

A map of robot sensors by what they detect



- **Contact** sensors: force/torque, tactile skins, limit switches.
- **Proximity** sensors: detect presence (yes/no) at short range.
- **Range** sensors: measure a distance (ultrasonic, LiDAR).
- Many give a simple **switching output** to a PLC input.

Strain gauges and the Wheatstone bridge





Foil strain gauge, Pleriche, CC BY-SA 4.0, via Wikimedia Commons

- A **strain gauge** is a thin foil resistor glued on metal: stretch it and R rises.
- Changes are tiny, so we use a **Wheatstone bridge** (left) to measure them.

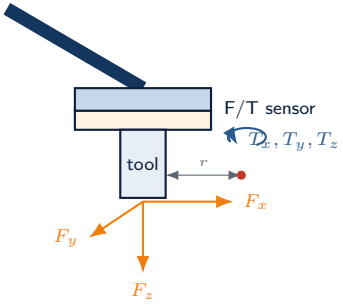
$$\frac{\Delta R}{R} = k_{\mathrm{GF}}\,\varepsilon \qquad (k_{\mathrm{GF}} \approx 2,\ \varepsilon = \text{strain} = \Delta L/L)$$

Worked example: full bridge

1. $\varepsilon = 10^{-3}$, $U_{\mathrm{ex}} = 5\,\mathrm{V}$, 4 active gauges
2. $u_{\mathrm{out}} \approx k_{\mathrm{GF}}\,\varepsilon\,U_{\mathrm{ex}} = 2 \cdot 10^{-3} \cdot 5\,\mathrm{V} = 10\,\mathrm{mV}$
3. Millivolts $\Rightarrow$ needs a low-noise amplifier

A **load cell** is a metal body with a bridge of gauges, rated in mV/V.

Six-axis force/torque sensor at the robot wrist



- Strain gauges on a stiff metal cross measure **3 forces + 3 torques**.
- Uses: force-controlled grinding, insertion, collision detection, hand guiding.

Refresher: torque

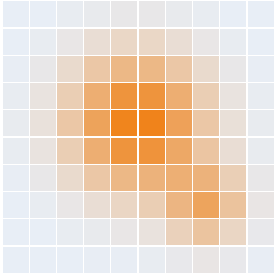
Torque = force × lever arm: $T = F\,r$ [N m].

Worked example: subtract the tool

1. Gripper mass 2 kg, centre of mass $r = 0.1$ m sideways
2. Weight $F_z = 2 \cdot 9.81 \approx 19.6$ N
3. Torque $T = 19.6 \cdot 0.1 \approx 1.96$ N m
4. The controller subtracts these (**gravity compensation**) so only *contact* forces remain.

The tool's weight changes direction as the wrist turns, so the compensation must use the current orientation.

Tactile sensors: a skin for the gripper



10×10 taxel pressure map

- A **tactile sensor** is an array of small pressure cells (**taxels**), like the nerve endings in a fingertip.
- Principles: resistive (pressure lowers R), capacitive (gap shrinks, C rises), piezoelectric (dynamic, slip).
- Gives **where** and **how hard** the object touches: detects slip, centres a grasp.

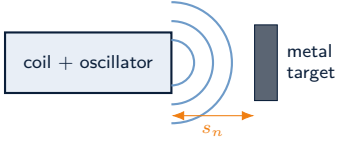
$$C = \varepsilon \frac{A}{d} \quad \Rightarrow \quad d \downarrow \quad C \uparrow$$

C capacitance [F], A plate area [m^2], d gap [m], ε permittivity [F/m].

Inductive and capacitive proximity sensors



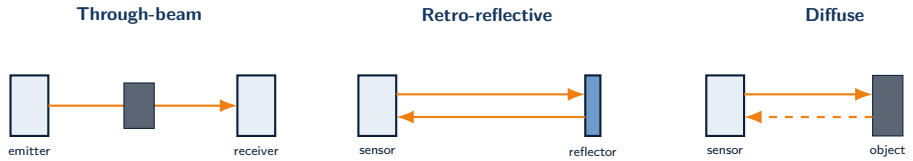
Inductive proximity switch, Lucasbosch, CC BY-SA 3.0, via Wikimedia Commons



- **Inductive**: an AC magnetic field; nearby **metal** absorbs energy (eddy currents) $\Rightarrow$ output switches.
- **Capacitive**: an electric field; any material (plastic, liquid, wood) changes the capacitance.
- **Sensing distance** s_n: a few mm; shrinks for non-steel metals.

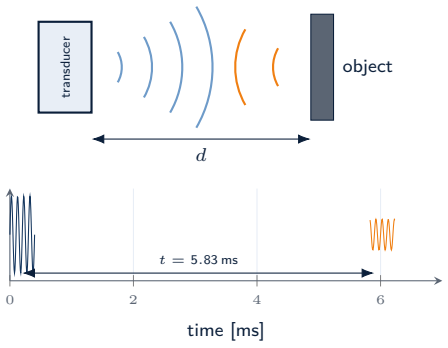
Target (inductive)	**factor on** s_n
mild steel	1.0
stainless steel	≈ 0.7
aluminium, copper	≈ 0.3–0.4

Photoelectric sensors: three ways to use a light beam



	Through-beam	Retro-reflective	Diffuse
Detects when	beam is **blocked**	beam is **blocked**	light **returns** from object
Range	longest (tens of m)	medium (several m)	short (cm to $\sim$1 m)
Wiring	two units	one unit + reflector	one unit
Weak spot	alignment	shiny objects	dark or far objects

Ultrasonic range sensors: listening for an echo



$$d = \frac{c\,t}{2}$$

c speed of sound (≈ 343 m/s at 20 °C), t echo time; divide by 2 for the round trip.

Worked example

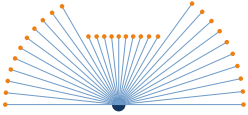
1. Echo after $t = 5.83$ ms
2. $d = 343 \cdot 0.00583/2 \approx 1.0$ m
3. At 40 °C, $c \approx 355$ m/s: same echo now means 1.035 m (3.5 % error if uncorrected).

- Works on glass and liquids; wide cone, slow, sensitive to temperature.

LiDAR and laser scanners: range with light



Velodyne HDL-64E LiDAR, Steve Jurvetson, CC BY 2.0, via Wikimedia Commons



2-D scan

- **LiDAR** (light detection and ranging): a laser pulse, the same $d = ct/2$, but $c = 3 \times 10^8$ m/s.
- A rotating mirror sweeps the beam $\Rightarrow$ a **point cloud** of distances.
- 2-D **safety laser scanners** guard robot cells (Session 5.7); 3-D LiDARs guide AMRs.

Worked example: nanoseconds

1. Object at 1 m: $t = 2d/c = 2/(3 \cdot 10^8) \approx 6.7$ ns
2. To resolve 1 cm we must time ≈ 67 ps
3. That is why LiDARs need very fast electronics, while ultrasound is easy to time.

Choosing a sensor for the task

Task in a robot cell	Good choice	Why
Is the steel part in the fixture?	inductive proximity	robust, cheap, metal only
Is the plastic bottle on the conveyor?	photoelectric (diffuse / retro)	any material, fast
Fill level of a bin of granules	ultrasonic	works on many surfaces
Insert a peg with 5 N contact force	6-axis F/T at the wrist	measures contact forces
Is the gripped object slipping?	tactile array	local pressure map
Person approaching the cell	safety laser scanner / light curtain	certified (PL d/e)
AMR navigation in a warehouse	LiDAR	accurate map of the room

Ask four questions: **what** material, **how far**, **yes/no or a value**, and **is it safety-related?** Safety functions need certified safety sensors, not standard ones.

Check your understanding: force, proximity and range

Check your understanding

1. An inductive sensor has $s_n = 8\,\text{mm}$ for steel. Roughly how close must an aluminium part be?
2. An ultrasonic echo returns after 2.9 ms. How far is the object (use 343 m/s)?
3. Why must a wrist F/T sensor "subtract the tool" before it can detect a collision?

Summary and key terms: force, proximity and range

- Strain gauges in a bridge $\rightarrow$ load cells and 6-axis F/T sensors.
- Tactile arrays give a pressure map for grasping.
- Inductive (metal), capacitive (any material), photoelectric (light beam).
- Range by time of flight: $d = ct/2$ (sound or light).
- Choose by material, distance, output type, safety.

Key terms

Wheatstone bridge measures tiny ΔR
F/T sensor 3 forces + 3 torques
taxel one tactile pressure cell
sensing distance s_n switching range
time of flight echo time to distance
point cloud set of measured 3-D points

[Fraden, 2016]; [Siciliano & Khatib, 2016]; IEC 60947-5-2.

Session 6

Day summary and outlook

Day summary

5.6 Conveyors, AGV/AMR, FMS, cobots: move parts, share space

5.7 ISO 12100 loop, PL_r, $S = KT + C$, stop categories

6.1 Measurement chain; accuracy, precision, resolution; calibration; τ

6.2 Encoders (incremental, absolute), resolver, IMU

6.3 F/T, tactile, inductive, capacitive, photoelectric, ultrasonic, LiDAR

Today in one sentence

A flexible, safe robot cell depends on sensors: they tell the robot where it is, where the parts are, and where the people are.

Three numbers to remember

$v = \pi D n / i$ conveyor speed

$S = K\,T + C$ safety distance

$d = c\,t/2$ time-of-flight range

Preview of tomorrow (W2D5): sensing, signals and integration

Code	Session
6.4	Machine vision in robotics
6.5	Sampling and digitisation
6.6	Signal processing and filters
6.7	Communication and integration
6.8	System-integration case study & wrap-up

Questions to think about

- How does a camera tell the robot where a part is?
- How often must we sample an encoder or a force sensor?
- How do 50 sensors and 6 drives share one network in time?

Figure credits

All plots and diagrams not listed here are original drawings made for this course (TikZ / pgfplots / circuitikz).

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- ISO 12100 (risk assessment and risk reduction); ISO 13849-1 (safety-related parts of control systems); IEC 62061.
- ISO 10218-1/-2 (industrial robots, robot systems and integration); ISO/TS 15066 (collaborative robots).
- ISO 13855 (positioning of safeguards); IEC 61496 (electro-sensitive protective equipment); IEC 60204-1 (electrical equipment of machines).
- ISO 9283 (robot performance criteria); IEC 60947-5-2 (proximity switches).